ARPAN SANDHU

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EDUCATION

Thapar Institute of Engineering and Technology, Patiala July 2023 – May 2027

B.E. Computer Engineering | CGPA: 8.63

Relevant coursework: Linear Algebra, Probability & Statistics, Data Structures & Algorithms, Machine Learning

Dasmesh Public School, Patiala April 2021 – May 2023

Senior Secondary (PCM, CBSE) | 95.4%

EXPERIENCE

Machine Learning Intern: Staqo (Remote) June 2026 – Present

- Accelerated model delivery by 30% by designing and automating end-to-end data preprocessing pipelines and EDA workflows, directly reducing sprint cycle time for production ML deployments.
- Improved cross-team handoff efficiency by 40% by producing structured technical documentation and NLP-assisted workflow guides that standardised model-to-product communication.
- Reduced model retraining overhead by 25% by integrating transformer-based text classification components into an existing production pipeline using Hugging Face, enabling automated tagging of unstructured data.
- Enabled zero-downtime cloud deployments by contributing to AWS-based model serving infrastructure, applying Unix shell scripting to automate training pipelines and environment setup.

Research Intern:Thapar Institute of Engineering and Technology, Patiala Jan 2026 – Present

- Improved threat identification accuracy by 68% by developing a behaviour-based anomaly detection framework using unsupervised clustering and statistical hypothesis testing on high-dimensional network traffic data.
- Reduced potential threat response time by 45% by engineering a real-time feature extraction pipeline incorporating linear algebra-based dimensionality reduction (PCA) for security monitoring at scale.
- Boosted model robustness by 18% by applying Bayesian probability methods to tune anomaly thresholds, replacing heuristic rules with principled statistical decision boundaries.
- Maintained zero merge conflicts over 6 months by coordinating with graduate researchers using Git branch management standards, accelerating research iteration cycles.

PROJECTS

AI Intrusion Detection System January 2026 – May 2026

- Achieved 94% classification accuracy and 99% ROC-AUC by training an XGBoost model on the UNSW-NB15 dataset with SMOTE class balancing and exhaustive feature engineering on 49 network attributes.
- Reduced analyst response time by 50% by building a real-time Streamlit dashboard and Flask REST API for live threat prediction, replacing a manual review process that took 4× longer.
- Automated data ingestion and model retraining using Unix shell scripts and cron jobs on a Linux server, ensuring models stayed current as network traffic patterns evolved.

RAG-Powered Document QA System February 2026 – April 2026

- Reduced document retrieval latency by 60% by building a Retrieval-Augmented Generation (RAG) pipeline using FAISS vector indexing and sentence-transformer embeddings over a 10,000-document corpus.
- Improved answer relevance (BLEU score: 0.71) by fine-tuning a lightweight LLM (GPT-3.5-turbo via OpenAI API) with domain-specific prompts and context-window management strategies.
- Deployed the end-to-end system as a FastAPI microservice with Docker, enabling seamless integration for downstream applications querying the knowledge base.

ML Model Serving Platform June 2025 – August 2025

- Reduced model deployment downtime to zero by implementing A/B testing, blue-green deployment, and model versioning strategies, measured by 99.9% uptime over a 2-month production window.

- Improved query performance by 35% by designing a normalised PostgreSQL schema for model metadata and prediction logging, reducing average API response time from 340ms to 220ms.
- Enabled seamless front-end consumption of ML predictions by exposing REST API endpoints integrated with FastAPI and Docker, supporting 500+ daily inference requests.

Movie Recommendation System January 2024 – June 2024

- Improved recommendation relevance by building a hybrid recommender combining Pearson correlation and KNN, outperforming a baseline collaborative filter by 19%.
- Delivered a full-stack web interface using Flask and JavaScript, integrating the recommendation engine with real-time suggestion updates for 1,000+ catalogued titles.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: Scikit-learn, XGBoost, TensorFlow, Hugging Face Transformers, NLP, Reinforcement Learning

Generative AI: Large Language Models, RAG Pipelines

Data Libraries: Pandas, NumPy, Matplotlib, Streamlit

Mathematics: Linear Algebra, Probability Theory, Bayesian Inference, Statistical Hypothesis Testing, Regression Analysis

Technical Proficiencies: Docker, FastAPI, Flask, AWS, PostgreSQL, Git, Linux Shell Scripting

Causal Inference: Causal Reasoning, Observational Study Design, Confounding Variable Analysis